

Skadi

Tartarini Luca
Patti Gabriele

June 15, 2026

Abstract

Due to global warming and rising temperatures, icebergs in the Arctic region are melting faster than ever, continuously changing their shape and location on the map, and this is often a problem for ships that navigate those seas. Today, to find the best route to navigate, the process is quite complicated: satellites scan the poles using SAR technology; as soon as they fly by an antenna, they download the image and send a huge amount of information to Ice Centers around the world. These Centers have to process it and post daily Ice Charts, to allow Arctic ships to plan their trip in advance, not considering continuous changes in the weather and in ice conditions, because updates can take up to 24 hours. Our solution consists of continuous tracking of ice conditions through SAR technology, implemented on a small swarm of satellites that orbit above the poles continually. Our satellites use on-board GPU to analyze raw radar data: it exploits a Convolutional Neural Network trained for semantic segmentation and classifies pixels into 'open water', 'first year ice' and 'multi-year ice'; it generates a vector map with icebergs contours, that are transmitted, straight to ships, via low-bandwidth links, and, finally, ship-borne computers calculate safe routes in real time. Our service can be distributed to the Arctic Coast Guard, to research centers and universities to study the variations in ice conditions, to insurance companies dealing with ships that navigate Arctic seas; to logistic companies that would be able to save millions of dollars by preventing iceberg-related damages, route delays and excess fuel consumption.

1 Introduction

Global trade is currently heavily reliant on, and increasingly vulnerable to, the fragility of traditional maritime chokepoints. Recent drought-induced restrictions in the Panama Canal and regional hostilities affecting the Suez Canal have dramatically underscored the necessity of alternative global shipping passages. The Arctic region offers a highly lucrative solution: for a vessel departing the Port of Taipei, the distance to New York is reduced from 10,842 nautical miles (NM) via the Panama Canal to 8,383 NM via the Northwest Passage (NWP)—a reduction of 2,459 NM. Similarly, the route from Taipei to Rotterdam via the Northern Sea Route (NSR) is only 8,101 NM, compared to 10,208

NM via the Suez Canal.

However, navigating these polar routes brings a unique set of challenges characterized by a phenomenon known as the 'Arctic Paradox'. While rising global temperatures are melting the polar ice caps and increasing the total number of navigable days each year, they simultaneously decrease the predictability of ice conditions. Consequently, while the Arctic is more accessible than ever, its navigability is always more dangerous.

Skadi offers a definitive solution to this problem utilizing a constellation of four medium-class satellites positioned in a Sun-Synchronous Dawn-Dusk orbit, ensuring near-continuous surveillance of the Arctic region. Equipped with C-band SAR technology, our satellites can penetrate cloud cover, operate in the dark of the polar winter, and distinctly classify ice types based on their structural properties.

2 Space Segment Architecture

2.1 Orbital Configuration

For our project we have chosen a Sun synchronous orbit (SSO) - inclination of 97.79° and altitude of 600km -, a polar orbit in which satellites match Earth's rotation around the Sun, in particular, in a dawn-dusk configuration, the satellites ride the terminator between day and night, so that they always face the Sun.

This specific orbit gives us several advantages, it allows our satellites to pass over the Arctic region on every orbit, roughly every 90 minutes, keeping our clients continuously updated; our solar panels are constantly illuminated, so that our SAR antenna is able to capture data during the entire duration of the Arctic overpass, also during the Arctic winter; avoiding thermal shocks, caused by entering or exiting the Earth's shadow, the satellites can stay at a constant temperature and reduce noise caused by thermal distortion; in this orbit, satellites pass over a certain region at the same time every day, ensuring our clients fixed shifts for ice conditions updates.

2.2 Ice Classification

Our technology to scan icebergs in the Arctic region, using microwave pulse physics, classifies them into "Open Water", "First-Year Ice", with a rough and salty surface, and "Multi-Year Ice", characterized by highly desalinated upper layers, resulting from suc-

cessive seasonal melt-freeze cycles, and a significant concentration of entrapped air bubbles. The waves emitted from our antenna are polarized in specific orientations that can be Horizontal (H) or Vertical (V); by combining these polarizations, we can obtain: Co-polarization (HH or VV), the radar transmits in one orientation and receives in the same orientation; or Cross-polarization (HV or VH), the radar transmits in one orientation but receives the orthogonal orientation. When the wave hits Open Water (OW), it reflects away, so the satellite does not receive the signal back, and, on the map, it results as black spots. First Year Ice (FYI) is still salty, so it conducts electricity, meaning that C-band microwaves cannot penetrate it deeply, the phenomenon of surface scattering makes the wave reflect in many directions, but with the same orientation, according to Co-polarization, partially going back to the satellite, and resulting in grey spots on the map. Multi-Year Ice (MYI) is full of air bubbles and mostly desalinated, so C-band microwaves can penetrate into the ice and bounce between the bubbles (volume scattering), resulting in a Cross-polarization and, finally, in bright spot on the map. Using this technique, we have a sort of double-check about the surface we are analyzing, reducing error percentage significantly.

Type	Characteristics and SAR signature
OW	Specular reflection (mirror-like, signal bounces away). SAR: Very dark in HH/VV, Black in HV/VH.
FYI	Surface scattering (bounces off rough, salty surface). SAR: Medium Grey in HH/VV, Dark Grey in HV/VH.
MYI	Volume scattering (signal penetrates desalinated ice, bounces off internal air bubbles). SAR: Bright in HH/VV, Uniquely Very Bright in HV/VH due to polarization scrambling

Table 1: Ice classification

2.3 Structure

A negative aspect of SSO is that one side of the satellite is always exposed to the Sun, while the other faces deep space, causing critical thermal issues. This configuration would bring the structure to reach extremely high temperatures on one side and deep space temperatures on the other side.

To deal with this problem, the side of our satellites that faces the Sun is covered with solar panels, to exploit as much power as possible, and made of several layers of MLI to prevent heat from reaching the core; the cold side is covered with radiators, to dissipate all the heat generated by the internal electronics into

space.

2.4 SAR Antenna

For our project we use a Synthetic Aperture radar (SAR) antenna, it works by sending discrete radar pulses to the ground, accumulating data from each pulse's echo, and then processing it through our on-board computer. A SAR antenna measures the range, brightness, and phase of each wavelength, giving a precise feedback of the analyzed surface. SAR has different imaging modes, like 'Spotlight' mode, the highest resolution mode; 'Dwell' mode, a very long spotlight mode; 'Strip' mode, a medium resolution mode, and 'Scan' mode, with modest resolution, but over large swath of land, in particular, we focused on ScanSAR, that captures a wide area via multiple consequent imaging, allowing us to control very wide surfaces like the Arctic region. Using this technique, we manage to reach 400km of swath, with a resolution of 20/25m, that allows us to track both icebergs and commercial ships. Many Earth observation companies use X-band antennas on their satellites, which allow them to capture high resolution images of the planet, but for our purpose, we have chosen C-band antennas with $\lambda = 5.5$ cm, which are capable of penetrating MYI and detecting air bubbles. We have tried to make it as compact as possible, and finally, with a length of 8m, we can ideally achieve an Azimuth resolution of 4m. To avoid possible disturbance echoes, its width is 1.2m.

Our antenna, since can't fit inside the satellite, has to exploit origami structure and advanced deployment mechanisms to be deployed and used. The success or not of our mission comes from this deployment mechanism, since if our antenna can't be deployed properly, we can't provide our service to the customers.

Parameter	Detail
Core Hardware	
Operating Frequency	C-Band ($\lambda \approx 5.5$ cm)
Dimensions	8 m (Az) $\times$ 1.2 m (El)
Technology	Active Electronically Scanned Array (AESA)
Peak Power	2,000 - 4,000 Watts
Modes & Performance	
ScanSAR	Res: 20 – 25 m Swath: 350 – 450 km
Revisit (NSR/NWP)	$\sim$ 24 minutes (4 satellites)
Glaciological Capabilities	
Polarization	Dual-Pol (HH / HV)
Ice Classification	MYI vs. FYI via HV channel

Table 2: C-Band SAR Antenna Specifications

2.5 Communication Architecture

Our GPU payload performs semantic segmentation directly in orbit, producing vector maps containing precise iceberg contours and ice classification polygons. This drastic reduction in data volume allows our satellites to bypass traditional ground relays for tactical data and utilize the VHF Data Exchange System (VDES), specifically its satellite component (VDE-SAT). VDE-SAT provides a bi-directional satellite communication channel specifically optimized for the exchange of small data payloads, such as meteorological updates, route planning, and ice condition maps. The primary commercial and engineering advantage of this architecture is its integration into existing maritime infrastructure. Arctic fleets are currently transitioning to standard VDES terminals; therefore, shipping companies are not required to install expensive, proprietary tracking antennas to receive our service. As the Skadi satellite overflies the targeted Arctic region, the operational sequence occurs within a single Line of Sight (LoS) window: the antenna scans the swath, the GPU immediately processes the data, and an omnidirectional VHF antenna broadcasts the vector map directly to the ship's VDES terminal. This guarantees Near Real-Time (NRT) tactical updates, maximizing navigational safety and delay avoidance.

2.6 AOCS

For our mission we have set 6 Operative Modes: Rate Reduction Mode (RRM) or detumbling, used after separation of the launcher, when the spacecraft has high angular rate; a phase of Sun pointing; a phase of antenna pointing, to capture data or to transmit them to the client; a phase of thrust vector pointing, to perform station keeping; and Safe Mode (SM), during which the satellite is kept with the solar arrays Sun-pointed. In order to collect data from the Arctic region and send them to the ships, we have to maintain an altitude of 600km with a tolerance of ± 1 km, this allow us to gather precise data of the region and making them easily understandable for the ships. We have chosen Hall-Effect Thrusters (HET) as main station-keeping actuators; this type of thrusters allow the satellite to keep constant thrust without excessive jittering drawback, is extremely efficient thanks to its high specific impulse ($I_{sp} = 1500 - 2000$ s), and can easily exploit the energy required by the antenna when SAR is not working. We have also implemented 4 reaction wheels (RW), one for each axis, and another for redundancy, for high precision maneuvers; and 3 magnetorquers, one for each axe, used to desaturate the excessive momentum of the wheels. For orbit determination we implemented a set of 3 star trackers (ST), ensuring a precise orientation even if one or two are temporarily blinded by the Sun or Earth's albedo; 1 Fine Sun Sensor (FSS), mounted on Sun-facing panel, and 4 Coarse Sun Sensors (CSS), mounted on the other panels, to orientate even during tumbling mode; a gyroscope to detect and correct even micro-vibrations; a magnetometer that measures

Earth's magnetic field, used with magnetorquers during detumbling phase; a GNSS receivers to track the satellite from the ground.

2.7 Electrical Power System (EPS)

The Electrical Power System is designed to support the demanding energy requirements of a high-performance C-band SAR antenna on a 1000 kg medium-class satellite. The 8m x 1.2m AESA antenna requires massive peak power (up to 4500 W during transmission).

To achieve this, Skadi relies on the properties of the Sun-Synchronous Dawn-Dusk orbit. By riding the Earth's terminator line, the satellite avoids Earth's shadow for the majority of the year, allowing for near-continuous solar illumination. The spacecraft is equipped with approximately 12 m² of deployable solar arrays utilizing high-efficiency triple-junction GaAs cells, capable of generating an average continuous power of 3000 W.

Even with this substantial generation, the SAR peak consumption exceeds the solar input, requiring the system to operate on a Duty Cycle. The SAR payload is active only during the Arctic overpass (approximately 15 minutes per 96-minute orbit). During this active phase, the power deficit is supplied by a high-capacity on-board Lithium-Ion battery pack.

Parameter	Value
Power Generation	
Solar Array Area	$\sim 12 \text{ m}^2$ (Deployable)
Average Power Generated	3000 W
Orbital Period	~ 96 minutes
Illumination Time	~ 96 minutes (No eclipse)
Power Consumption & Storage	
SAR Active Mode (Peak)	5000 W (SAR + Bus)
SAR Duty Cycle	$\sim 15 \text{ min}$ / orbit
Energy per SAR pass	$\sim 1250 \text{ Wh}$
Recharge Mode (Bus + HET + GPU)	800 W
Battery Specs	
Battery Type	Lithium-Ion (Li-Ion)
Total Capacity	5.0 kWh
Max Depth of Discharge (DoD)	25% (For 5-7 years life)

Table 3: EPS Power Budget and Duty Cycle

During the remaining 81 minutes of the orbit, the SAR payload is placed in standby. The 3000 W generated by the solar panels easily covers the 800 W required by the satellite bus, GPU edge-computing processing, downlinking, and HET station-keeping maneuvers. The remaining 2200 W of excess power is used to fully recharge the battery pack before the next Arctic overpass. To guarantee a 5 to 7-year mission lifespan in Low Earth Orbit without severe degradation, the

battery pack is sized to 5.0 kWh, ensuring the Depth of Discharge (DoD) never exceeds 25% per orbit.

3 On-Board Intelligence and AI

Traditional Earth observation missions follow a bent-pipe paradigm: raw SAR data are stored on-board, downlinked when a ground station is in sight, and subsequently processed by Ice Centers on the ground. A single ScanSAR acquisition over the Arctic generates between 2 and 4 GB of complex I/Q data, and the polar geometry means that a suitable ground contact may not be available for several hours after acquisition. This latency is structurally incompatible with the Near Real-Time updates that Arctic navigation requires. Skadi addresses this constraint by relocating the processing chain from the ground to the satellite itself. Each spacecraft carries an on-board edge-computing payload, GPU, designed to perform image formation, semantic segmentation, and vectorization directly in orbit. The output is not a radar image but a compact vector map encoding ice-type polygons and iceberg contours, small enough to be transmitted over a low-bandwidth VDE-SAT link within the same Line-of-Sight window in which the data were acquired. The expected compression ratio between raw SAR input and vector output is on the order of $1000\times$, which is the enabling condition for the communication architecture described in Section 2.5.

3.1 The GPU Edge-Computing Payload

GPU is an FPGA-based accelerator and qualified for the radiation environment of LEO. The platform is built around a Xilinx Zynq UltraScale+ MPSoC, which combines a quad-core ARM Cortex-A53 processor, responsible for pipeline control and scheduling, with a reconfigurable logic fabric that implements the convolutional layers of the neural network as dedicated hardware datapaths. This approach avoids the overhead of a general-purpose processor running inference through a software stack and keeps both latency and power consumption within the constraints of a medium-class satellite bus.

Two properties of this architecture are particularly relevant to a safety-critical application. First, the design incorporates SEU mitigation techniques and Triple Modular Redundancy on critical registers, reducing the probability that a cosmic-ray hit produces a silent error in the classification output. Second, the platform operates within a power envelope below 30 W, which is negligible relative to the 4,000–5,000 W peak demand of the SAR antenna and compatible with the 800 W bus budget during the non-acquisition phase described in Section 2.6.

The accelerator supports INT8 and FP16 quantized inference natively, which is a necessary condition for running a convolutional neural network within the on-chip memory of the Zynq fabric without relying on exter-

nal DRAM accesses, which would otherwise dominate both latency and power draw. The company that produces our GPU provides complementary space-grade software IP for hardware abstraction and task scheduling, reducing integration complexity for the on-board software team.

3.2 CNN Architecture for Sea-Ice Segmentation

The segmentation task consists of assigning each pixel in the processed SAR image to one of three classes: Open Water (OW), First-Year Ice (FYI), or Multi-Year Ice (MYI). This is a standard dense prediction problem, and the intended architecture is a U-Net with a MobileNetV3-Small encoder.

The U-Net structure is selected for two reasons specific to this application. Its skip connections between encoder and decoder stages retain the spatial detail needed to reconstruct iceberg boundaries at the resolution of the ScanSAR product (20–25 m). At the same time, the MobileNetV3-Small backbone uses depthwise-separable convolutions, which reduce the multiply-accumulate count significantly compared to a standard convolutional encoder, keeping the total parameter count in the range of approximately 5 million — small enough, once quantized to INT8, to reside entirely within the on-chip BRAM and URAM of the Zynq fabric.

The network input is a two-channel tile of 512×512 pixels, one channel carrying HH intensity and one carrying HV intensity. The motivation for this dual-channel design follows directly from the scattering physics described in Section 2.2: the distinction between MYI and FYI is most reliably encoded in the relationship between co-polarized and cross-polarized backscatter, so providing both channels as separate inputs allows the model to exploit this physical contrast rather than having to infer it from a merged single-channel representation.

3.3 Training Dataset and Strategy

The intended primary training corpus is the AI4Arctic Sea Ice Challenge dataset, a publicly available collection of co-located Sentinel-1 dual-polarimetric C-band SAR acquisitions and ice chart annotations produced by the Danish Meteorological Institute (DMI) and the Canadian Ice Service (CIS). The dataset is appropriate for this application because the radar configuration — C-band, HH and HV polarizations, acquired over Arctic waters — matches our own antenna design closely, limiting the domain gap between training data and operational acquisitions.

The base dataset is planned to be supplemented with Sentinel-1 scenes co-registered with ice charts from the U.S. National Ice Center (NIC), extending seasonal and geographic coverage, and with a set of synthetic augmentations including speckle noise injection, incidence-angle-dependent intensity scaling, and geometric transformations, intended to improve robust-

ness to the varying acquisition geometries of ScanSAR mode.

The planned loss function combines weighted categorical Cross-Entropy with a Dice loss term. Cross-Entropy manages the class imbalance — Open Water occupies the majority of Arctic scenes by area — while the Dice term optimizes region-level overlap, which is more informative than per-pixel accuracy under imbalanced conditions. Class weights in the loss are intended to penalize MYI false negatives more heavily than OW errors, reflecting the asymmetric cost of missing the most hazardous ice type relative to a false alarm on open water.

Following convergence, the network weights will undergo Quantization-Aware Training (QAT) to recover the accuracy partially lost in the transition from FP32 to INT8 before the weights are uploaded to GPU flash memory.

3.4 On-Board Processing Pipeline

After each ScanSAR acquisition, the GPU executes a five-stage pipeline designed to complete within the approximately 15-minute Arctic overpass window.

1. **Image Formation.** Raw I/Q echoes are focused into a Single-Look Complex (SLC) image using a Range-Doppler algorithm implemented in the FPGA fabric. The parallel architecture of the re-configurable logic is well suited to the fast Fourier transforms and range compression operations that this step requires.
2. **Radiometric Calibration and Speckle Filtering.** A Lee filter is applied to reduce the multiplicative speckle noise intrinsic to coherent radar imaging. Pixel values are then converted to normalized backscatter coefficients σ^0 , placing HH and HV on a consistent radiometric scale across the swath.
3. **Tiling and Inference.** The calibrated image is partitioned into overlapping 512×512 tiles, which are processed in batch by the U-Net to produce a per-pixel class probability map. Overlapping regions are merged using a weighted blending function to avoid discontinuities at tile boundaries.
4. **CFAR Iceberg Detection.** A Constant False Alarm Rate (CFAR) detector is applied in parallel on the HH channel to identify isolated bright point targets, primarily small icebergs and growlers, whose spatial extent falls below the segmentation grid resolution. These targets are maintained as a separate point catalogue and transmitted alongside the polygon layer, since they represent a navigational hazard that the segmentation mask alone would not capture.
5. **Vectorization and Compression.** The raster segmentation mask is converted to closed polygons via contour extraction and simplified using the Douglas-Peucker algorithm at a tolerance matched

to the ScanSAR ground resolution. The resulting payload — class-labeled polygons with associated confidence scores and the CFAR point catalogue — is encoded in a compact binary format and passed to the VDE-SAT transmitter. The expected output size is in the range of 1–3 MB per swath.

The full pipeline is estimated to require approximately 40 seconds per swath, which fits comfortably within the overpass window.

3.5 Robustness and Updateability

Three mechanisms are planned to address reliability and long-term maintainability.

A per-tile confidence score, derived from the entropy of the softmax output distribution, is attached to each polygon in the transmitted map. Polygons whose confidence falls below a defined threshold are flagged as uncertain, allowing the on-bridge navigation software to distinguish high-confidence classifications from cases where the model output should be treated with caution.

An out-of-distribution detection module, based on input image statistics, runs alongside the segmentation pipeline. If the statistical properties of an acquired scene differ substantially from the training distribution — indicating an unusual atmospheric condition or an ice formation not well represented in the training data — the frame is flagged and a re-acquisition is scheduled on the following overpass rather than transmitting a classification of unknown reliability.

Finally, the CNN weights are stored in a reprogrammable memory partition on the GPU. This allows updated model versions to be uplinked from the ground segment throughout the mission lifetime as the training dataset expands, as the model is refined with operational data, or as ice regime characteristics evolve, ensuring that segmentation performance is not fixed at launch.

4 Ground and Maritime Infrastructure

The Skadi constellation generates actionable ice intelligence in orbit, but the operational value of this data depends on the architecture through which it is distributed. The infrastructure is divided into two paths: the space-to-ground link, intended to serve administrative, archival, and commercial functions; and the direct space-to-ship link, delivering Near Real-Time (NRT) tactical data to the vessel.

4.1 Ground Segment and Cloud Platform

Ground stations are intended to serve as the primary interface for constellation command and control. They would receive telemetry, health logs, and diagnostic

Parameter	Value
Edge-Computing Hardware	
Platform	GPU
SoC	Xilinx Zynq UltraScale+ MPSoC
Power Envelope	< 30 W
Radiation Hardening	SEU mitigation + TMR
Inference Precision	INT8 / FP16
Neural Network	
Architecture	U-Net (MobileNetV3-Small encoder)
Input	$512 \times 512 \times 2$ (HH, HV)
Output Classes	OW, FYI, MYI
Parameter Count	~ 5 M (INT8 quantized)
Training Dataset	AI4Arctic (DMI / CIS)
Loss Function	Cross-Entropy + Dice (weighted)
Pipeline Performance	
Raw SAR Input	2–4 GB per swath
Pipeline Latency	~ 40 s per swath
Vector Map Output	1–3 MB per swath
Compression Ratio	$\sim 1000\times$

Table 4: On-Board AI Payload Specifications

data from the four satellites, and uplink operational commands, orbital maneuver parameters, and updated AI model weights as described in Section 3.5, with ground contacts scheduled at regular intervals to ensure continuous oversight of the constellation’s state. The ground station infrastructure is planned to be procured as a managed service from established providers of Ground Station as a Service (GSaaS).

From the ground station, data would be routed via secure terrestrial links to the Skadi Cloud Platform, hosted on European cloud infrastructure. This platform is intended to host the Mission Control Center (MCC) software, which manages satellite scheduling, orbit determination, payload configuration, and anomaly resolution.

The cloud platform is also designed to act as the central data repository. All vector maps transmitted to ships would be simultaneously archived in a persistent database. This repository would power the Skadi Web Portal, a browser-based interface for viewing current and historical ice maps, and an external REST API through which authorized clients can query and download data programmatically. These interfaces support the Data-as-a-Service (DaaS) model described in Section 5: potential users include government agencies, coast guards, research institutions, and insurance companies, each accessing the data relevant to their operational or analytical needs.

4.2 The SkadiBox

On the maritime side, the distribution architecture is designed to minimize integration effort for the end user. Modern Arctic vessels are already equipped with VDES antennas, modems, and an internal Local Area Network (LAN), operating over either wired Ethernet or Wi-Fi. Skadi is intended to leverage this existing hardware, so that shipping companies are not required to install proprietary tracking antennas or specialized communication equipment to receive the service.

The only planned hardware addition is the SkadiBox, a ruggedized, low-power compute unit that connects directly to the ship’s LAN. The SkadiBox would interface with the vessel’s existing VDES modem to receive the incoming vector map broadcasts from the overflying satellites.

Once a vector map is received, the SkadiBox would decode the binary payload and store it on an internal solid-state drive. This local storage is intended to maintain a rolling archive of past ice conditions along the vessel’s corridor, allowing navigators to review historical data and observe trend evolution even when satellite contact is temporarily unavailable.

The SkadiBox is designed to expose the ice data to the bridge through two local interfaces. A lightweight web application, accessible from any standard browser on the ship’s LAN, would provide a visual overlay of ice-type polygons, iceberg point targets from the CFAR catalogue, and per-polygon confidence flags on a nautical chart background. In addition, a local REST API would allow the vector data to be ingested directly by the ship’s Electronic Chart Display and Information System (ECDIS) or by third-party route optimization software. This approach ensures that Skadi’s data enters the navigator’s existing workflow without requiring separate display hardware on the bridge.

5 Business Idea

5.1 Cap-Ex

For the production of each satellite, we have estimated a cost between \$35M and \$40M, resulting in a total Capex of \$140M - \$160M, for a swarm of 4 satellites. This cost comes mostly from the SAR payload, for which we have estimated a \$20M expense, divided between the antenna and the deployment mechanism; while the rest comes from the bus and Assembly, Integration, and Testing (AIT). For R&D, in order to produce an advanced and failure-proof technology, we have planned an expense of $\sim$ \$30M. For the ground segment, we have estimated a cost of \$10M - \$15M, considering the acquisition of radio frequencies authorization, and the cloud infrastructure and Mission Control Center development.

Moreover, for launch and insurance we have estimated a cost of \$40M - \$60M.

CAPEX Component	Cost
Satellite Production	\$140M - \$160M
R&D	~ \$30M
Launch and Insurance	\$40M - \$60M
Ground segment	\$10M - \$15M
Total CAPEX	\$220M - \$265M

Table 5: CAPEX Summary

5.2 Op-Ex

Our operating capital is significantly reduced thanks to GPU edge computing, indeed this technology allows us to transmit light vector files instead of massive raw data, in an estimated annual expense of \$1.0M - \$1.5M. Other annual expenses considered are human capital, cloud computing and storage to sustain our infrastructure, licenses and insurance for an estimated annual Opex of \$8.0M - \$12M.

OPEX Component	Cost
Ground Sattion Down-link	\$1.0M - \$1.5M
Cloud Computing and Storage	\$1.0M - \$2.0M
Human Capital	\$4.5M - \$6.0M
Insurance and Licenses	\$1.5M - \$2.5M
Total Annual OPEX	\$8.0M - \$12M

Table 6: OPEX Summary

5.3 Business Plan

The main customers of our project are, obviously, the companies that navigate the Arctic sea for tourism and commercial purposes.

For a specialized cargo ship (Arc7) the average costs of navigation are \$120K - \$130K per day, so \$5000 - \$5400 per hour, these include fuel consumption needed to overcome ice resistance, insurance and crew costs. Getting stuck in the ice means having to keep the engines running at a higher RPM than normal just to power the anti-freezing and life-support systems, consuming significantly more fuel and, moreover, commercial penalties for the delayed delivery of the cargo. From late July to early October delays are relatively rare, estimated to be 4% of total voyages, and they last usually 1-3 days.

The maximum risk is in May, June, October and November, when ice can melt late or reform early. While delays of 1-3 days are almost the norm, due to fog speed reduction and deviations, prolonged blockages or waits for icebreakers are also very frequent, and can bring to delays of over 8 days; these meteorological crises happen about 1 or 2 times a year and can involve many ships, like in November 2021, when 24 ships were trapped in the ice waiting for icebreakers. With our service we aim to reduce these delays of 1-2

Company	Fleet (ships)
Sovcomflot (Russia)	83
MOL (Japan)	130
Seapeak (USA/Canada)	94
Dynagas (Greece)	26
Royal Wagenborg (The Netherlands)	140
COSCO (China)	480
Fednav (Canada)	120
Nordic Bulk Carriers (Denmark)	11
NEAS (Canada)	5
Fesco (Russia)	30
Ponant (France)	13
Quark Expeditions (USA/Canada)	7
Hurtigruten/HX (Norway)	15
Lindblad Expeditions (USA/Canada)	22
Silversea (Monaco)	13
Hapag-Lloyd Cruises (Germany)	5
Western Alliance's fleet	601
European countries' fleet	223

Table 7: Arctic companies' fleets

days, resulting in a saving of \$120K - \$260K per trip, excluding the savings from the 'delay avoidance'.

We make deals with companies that need our service to improve their commercial activities in the region, and can perform several deliveries per year; our fee is \$300K per ship, with the requirement to implement the service on the whole fleet.

Besides direct sales to shipping companies, insurance firms represent a highly lucrative secondary market. By providing historical data for premium calculation and real-time monitoring for loss prevention, Skadi mitigates multi-million dollar claims related to hull damage or ice-trapping. We propose an Enterprise Data Subscription model for insurers, priced between \$500K and \$1.5M annually.

Finally, our data have a strong geopolitical power, and can easily sold with a Data-as-a-Service (DaaS) model to institutions and research centers.

With an expected revenue of ~ 80M per year (we have considered 223 ships to use Skadi), we would reach the break-even point after 3 years and 325 days from the launch.

5.4 Go-to-market Strategy

Our go-to-market strategy consists in analyzing data from governative satellites using our artificial intelligence to classify the ice from ground, and sending the maps to the ships in the Arctic region. This strategy, despite doesn't respect our Near Real Time updates criteria, allows us to validate our technology, by testing it on client ships and receiving suggestions from the customers to improve our service.

6 Social Impact

The opening of the Northwest Passage (NWP) and Northern Sea Route (NSR) is redefining global maritime trade. By making these routes reliably navigable, Skadi reduces global dependence on traditional, politically sensitive chokepoints like the Suez and Panama canals. This diversification of trade routes enhances global supply chain resilience, ensuring the stable flow of essential goods, food, and energy resources even amidst regional conflicts, piracy threats, or climate-induced infrastructure failures at lower latitudes. Operating in the Arctic, however, is inherently dangerous. Historical data shows that 26% of iceberg collisions result in the vessel sinking. Current Arctic insurance premiums are up to three times higher than conventional routes due to this risk to human life and cargo. Skadi aims to mitigate these risks by providing Near Real-Time (NRT) tactical data directly to the ship's bridge. By anticipating iceberg movements and accurately classifying ice thickness, we drastically reduce the probability of collisions, structural damage, and shipwrecks, fundamentally safeguarding the lives of the crews operating in these extreme environments where rescue operations are highly complex. Furthermore, increasing maritime traffic in the poles presents a severe environmental paradox. While shorter transits between Asia and Europe via the Arctic reduce global greenhouse gas emissions by cutting voyage distances by up to 40%, an accident in these pristine waters would be ecologically catastrophic. An oil spill or fuel leak in the Arctic is exceptionally difficult to clean up due to the rapid formation of ice, lack of infrastructure, and harsh weather, posing an existential threat to local marine biodiversity. By significantly reducing the likelihood of vessels breaching their hulls against MYI, Skadi acts as an active environmental shield. Additionally, by optimizing routes, ships avoid the excessive fuel consumption and black carbon emissions associated with forcing their way through unexpected ice ridges or keeping engines running at high RPMs while trapped waiting for icebreakers. Finally, safer and more predictable navigation has a direct, positive impact on local populations. It ensures the reliable and cost-effective delivery of critical supplies to remote Arctic and Indigenous communities, which heavily rely on maritime transport for fuel, food, and medical supplies during the short and unpredictable navigable season. The Arctic is rapidly transitioning from a frozen frontier into a highly contested geopolitical arena. As sea ice retreats, the Northern Sea Route (NSR) and the Northwest Passage (NWP) are becoming viable alternatives to traditional transit corridors. However, these routes are the subject of ongoing territorial disputes: Russia and Canada claim the NSR and NWP, respectively, as internal sovereign waters, while the United States and the European Union maintain they are international straits subject to the right of transit passage.

7 Geopolitical Implications

In this context, independent data is a strategic asset. Currently, navigating the NSR heavily relies on data, pilotage, and icebreaker escorts provided by Russian state-owned enterprises, such as Rosatomflot. This creates a severe strategic vulnerability and an information monopoly. By deploying an autonomous, high-resolution Earth observation constellation, Skadi provides Western and allied shipping companies with independent, third-party situational awareness. This empowers operators to navigate Arctic waters with reduced dependency on state-monopolized infrastructure, thereby supporting the principle of freedom of navigation and enhancing European and Western strategic autonomy. Furthermore, SAR technology is inherently dual-use. While Skadi is primarily designed for commercial route optimization, the continuous, all-weather surveillance of the Arctic provides critical Maritime Domain Awareness (MDA) for allied defense and intelligence agencies. The ability to monitor "dark vessels" (ships navigating with their Automatic Identification System disabled), unauthorized resource extraction, and the rapid expansion of military infrastructure in the high North makes our data exceptionally valuable. Consequently, our Data-as-a-Service (DaaS) model extends beyond commercial logistics, offering a crucial intelligence tool for Coast Guards, navies, and allied governments seeking to monitor and secure their northern geopolitical interests.

References

- [1] W. Aldenhoff, L. E. B. Eriksson, Y. Ye, and C. Heuze. First-year and multiyear sea ice incidence angle normalization of dual-polarized sentinel-1 sar images in the beaufort sea. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 13:1540–1550, 2020.
- [2] M. Arya et al. Ultralight deployable space structures. *Aerospace Research Central*, 2017.
- [3] H. D. Curtis. *Orbital Mechanics for Engineering Students*. Butterworth-Heinemann, 2013.
- [4] L. Diana and P. Dini. Review on hardware devices and software techniques enabling neural network inference onboard satellites. *Remote Sensing*, 16(21):3957, 2024.
- [5] W. Dierking. Sea ice monitoring by synthetic aperture radar. *Oceanography*, 26(2):100–111, 2013.
- [6] V. M. Eguíluz et al. A quantitative assessment of arctic shipping in 2010–2014. *Scientific Reports*, 6:30682, 2016.
- [7] D. G. Gilmore. *Spacecraft Thermal Control Handbook, Volume I: Fundamental Technologies*. The Aerospace Press, 2002.

- [8] D. M. Goebel and I. Katz. *Fundamentals of Electric Propulsion: Ion and Hall Thrusters*. JPL Space Science and Technology Series / NASA, 2008.
- [9] G. Halpert et al. Aerospace applications of lithium-ion batteries. *Journal of Power Sources*, 80(1–2):262–266, 1999.
- [10] M. Hoefft, K. Gierlowski, J. Rak, J. Wozniak, and K. Nowicki. Non-satellite broadband maritime communications for e-navigation services. *IEEE Access*, 9:62697–62718, 2021.
- [11] ICEYE. Beginner’s guide to synthetic aperture radar (sar) technology.
- [12] Rajesh Kandel and Hiba Baroud. A data-driven risk assessment of arctic maritime incidents: Using machine learning to predict incident types and identify risk factors. *Reliability Engineering System Safety*, 243:109779, 2024.
- [13] F. Lasserre. Simulating shipping along arctic routes: The profitability of the northern sea route and the northwest passage. *Maritime Policy & Management*, 2014.
- [14] T. Nilsen. Crisis on northern sea route as early freeze traps dozens of ships. *The Independent Barents Observer*, 2021.
- [15] R. G. Onstott. *SAR and Scatterometer Signatures of Sea Ice*. 1992.
- [16] PAME (Protection of the Arctic Marine Environment). Arctic shipping status reports (assr). Technical report, Arctic Council, 2023.
- [17] M. R. Patel. *Spacecraft Power Systems*. CRC Press, 2004.
- [18] H. Schøyen and S. Bråthen. The northern sea route versus the suez canal: cases from bulk shipping. *Journal of Transport Geography*, 19(4):977–983, 2011.
- [19] M. Shokr and M. Dabboor. Polarimetric sar applications of sea ice: A review. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 16:6627–6641, 2023.
- [20] The Barents Observer and High North News. Sudden freeze-up disrupts supply chain on russia’s northern sea route. Early Winter Freeze Traps Ships in Arctic Ice.
- [21] R. Torres et al. Gmes sentinel-1 mission. *Remote Sensing of Environment*, 120:9–24, 2012.
- [22] R. van den Broek. The economic impact of open arctic routes on global maritime lng trade. Technical report. Cross-referenced with the financial reports of Arctic operators like Teekay LNG and Dynagas.
- [23] H. Wan, X. Luo, Z. Wu, X. Qin, X. Chen, B. Li, J. Shang, and D. Zhao. Multi-featured sea ice classification with sar image based on convolutional neural network. *Remote Sensing*, 15(16):4014, 2023.
- [24] J. R. Wertz, D. F. Everett, and J. J. Puschell. *Space Mission Engineering: The New SMAD*. Microcosm Press, 2011. Capitolo 20 sulle Power Systems.
- [25] V. Y. Zakhvatkina et al. Classification of sea ice types in envisat asar images. *IEEE Transactions on Geoscience and Remote Sensing*, 51(5):2587–2600, 2013.